MSG No. 122.338 $I_c\bar{4}2d$ [Type IV, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `-4..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,4,5,6]`
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,15,16]`
5	$\left[0, 0, \frac{1}{2}\right]$	`[17,20,21,22]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[18,19,23,24]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,28,29,30]`
8	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[26,27,31,32]`

Table 2: Wyckoff site: `8b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,21,22]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,3,23,24]`
3	$\left[0, 0, \frac{3}{4}\right]$	`[5,6,17,20]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,18,19]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,12,29,30]`
6	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,11,31,32]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[13,14,25,28]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[15,16,26,27]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'22'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{8}\right]$	`[1,2,27,28]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{8}\right]$	`[3,4,25,26]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{7}{8}\right]$	`[5,15,24,30]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{7}{8}\right]$	`[6,16,23,29]`
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{8}\right]$	`[7,13,22,32]`
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{8}\right]$	`[8,14,21,31]`
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{5}{8}\right]$	`[9,10,19,20]`
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{5}{8}\right]$	`[11,12,17,18]`

Table 4: Wyckoff site: `8d`, site symmetry: `2'2'2.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{8}\right]$	`[1,11,18,28]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{7}{8}\right]$	`[2,12,17,27]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{7}{8}\right]$	`[3,9,20,26]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{8}\right]$	`[4,10,19,25]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{5}{8}\right]$	`[5,8,30,31]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{5}{8}\right]$	[6,7,29,32]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{8}\right]$	[13,16,22,23]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right]$	[14,15,21,24]

Table 5: Wyckoff site: 16e, site symmetry: 2..

No.	position	mapping
1	$[0, 0, z]$	[1,4]
2	$\left[0, \frac{1}{2}, \frac{1}{4} - z\right]$	[2,3]
3	$[0, 0, -z]$	[5,6]
4	$\left[0, \frac{1}{2}, z + \frac{1}{4}\right]$	[7,8]
5	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[9,12]
6	$\left[\frac{1}{2}, 0, \frac{3}{4} - z\right]$	[10,11]
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[13,14]
8	$\left[\frac{1}{2}, 0, z + \frac{3}{4}\right]$	[15,16]
9	$\left[0, 0, z + \frac{1}{2}\right]$	[17,20]
10	$\left[0, \frac{1}{2}, \frac{3}{4} - z\right]$	[18,19]
11	$\left[0, 0, \frac{1}{2} - z\right]$	[21,22]
12	$\left[0, \frac{1}{2}, z + \frac{3}{4}\right]$	[23,24]
13	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[25,28]
14	$\left[\frac{1}{2}, 0, \frac{1}{4} - z\right]$	[26,27]
15	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[29,30]
16	$\left[\frac{1}{2}, 0, z + \frac{1}{4}\right]$	[31,32]

Table 6: Wyckoff site: 16f, site symmetry: .2.

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{8}\right]$	[1,2]
2	$\left[-x, \frac{3}{4}, \frac{1}{8}\right]$	[3,4]
3	$\left[\frac{1}{4}, -x, \frac{7}{8}\right]$	[5,15]
4	$\left[\frac{3}{4}, x, \frac{7}{8}\right]$	[6,16]
5	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{3}{8}\right]$	[7,13]
6	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{3}{8}\right]$	[8,14]
7	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	[9,10]
8	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{5}{8}\right]$	[11,12]
9	$\left[x, \frac{1}{4}, \frac{5}{8}\right]$	[17,18]
10	$\left[-x, \frac{3}{4}, \frac{5}{8}\right]$	[19,20]
11	$\left[\frac{1}{4}, -x, \frac{3}{8}\right]$	[21,31]
12	$\left[\frac{3}{4}, x, \frac{3}{8}\right]$	[22,32]
13	$\left[\frac{3}{4}, \frac{1}{2} - x, \frac{7}{8}\right]$	[23,29]
14	$\left[\frac{1}{4}, x + \frac{1}{2}, \frac{7}{8}\right]$	[24,30]
15	$\left[x + \frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	[25,26]
16	$\left[\frac{1}{2} - x, \frac{1}{4}, \frac{1}{8}\right]$	[27,28]

Table 7: Wyckoff site: `16g`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{3}{4}, y, \frac{1}{8}\right]$	`[1,27]`
2	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{8}\right]$	`[2,28]`
3	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{8}\right]$	`[3,25]`
4	$\left[\frac{1}{4}, -y, \frac{1}{8}\right]$	`[4,26]`
5	$\left[y, \frac{1}{4}, \frac{7}{8}\right]$	`[5,24]`
6	$\left[-y, \frac{3}{4}, \frac{7}{8}\right]$	`[6,23]`
7	$\left[-y, \frac{3}{4}, \frac{3}{8}\right]$	`[7,22]`
8	$\left[y, \frac{1}{4}, \frac{3}{8}\right]$	`[8,21]`
9	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{5}{8}\right]$	`[9,19]`
10	$\left[\frac{1}{4}, -y, \frac{5}{8}\right]$	`[10,20]`
11	$\left[\frac{3}{4}, y, \frac{5}{8}\right]$	`[11,17]`
12	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{5}{8}\right]$	`[12,18]`
13	$\left[y+\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[13,32]`
14	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{3}{8}\right]$	`[14,31]`
15	$\left[\frac{1}{2}-y, \frac{1}{4}, \frac{7}{8}\right]$	`[15,30]`
16	$\left[y+\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	`[16,29]`

Table 8: Wyckoff site: `16h`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,28]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}-z\right]$	`[2,27]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[3,26]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[4,25]`
5	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[5,30]`
6	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[6,29]`
7	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{4}\right]$	`[7,32]`
8	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[8,31]`
9	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[9,20]`
10	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}-z\right]$	`[10,19]`
11	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-z\right]$	`[11,18]`
12	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[12,17]`
13	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[13,22]`
14	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[14,21]`
15	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[15,24]`
16	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{3}{4}\right]$	`[16,23]`

Table 9: Wyckoff site: `32i`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[x, \frac{1}{2}-y, \frac{1}{4}-z\right]$	`[2]`
3	$\left[-x, y+\frac{1}{2}, \frac{1}{4}-z\right]$	`[3]`
4	$[-x, -y, z]$	`[4]`

continued ...

Table 9

No.	position	mapping
5	$[y,\ -x,\ -z]$	[5]
6	$[-y,\ x,\ -z]$	[6]
7	$\left[-y,\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	[7]
8	$\left[y,\ x+\frac{1}{2},\ z+\frac{1}{4}\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-y,\ -x,\ z+\frac{3}{4}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x,\ z+\frac{3}{4}\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[x,\ \frac{1}{2}-y,\ \frac{3}{4}-z\right]$	[18]
19	$\left[-x,\ y+\frac{1}{2},\ \frac{3}{4}-z\right]$	[19]
20	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[22]
23	$\left[-y,\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	[23]
24	$\left[y,\ x+\frac{1}{2},\ z+\frac{3}{4}\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{4}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{4}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[28]
29	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[29]
30	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ -z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ -x,\ z+\frac{1}{4}\right]$	[31]
32	$\left[y+\frac{1}{2},\ x,\ z+\frac{1}{4}\right]$	[32]